

CJ Nour

Electrical/Software Engineer
Residing in Etobicoke, Ontario

Portfolio: cjnour.com
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SKILLS

Electrical: EPLAN, PCB Design, Soldering, Servo Motor Control, PLC Programming
Programming: Java, React.js, Python, C++, IoT, Git, Linux
CAD: NX, Autodesk Inventor, EPLAN, 3D Printing & Prototyping
Analysis: Excel, MATLAB/Simulink, Root Cause Analysis, Hardware-in-the-Loop Testing

PROFESSIONAL EXPERIENCE

Husky Technologies | Electrical Engineer Jun. 2025 – Present
EPLAN, NX, Servo Motors, UL/EN, DFM, DFA, Root Cause Analysis Bolton, Ontario

- Engineered electrical cabinets for 3 next-generation prototypes of a best-selling machine using EPLAN and NX. Applied DFM/DFA principles to optimize safety circuits, and fuse / circuit breaker sizing to comply with UL 508A and EN standards.
- Led a root-cause analysis to revamp a complex calculator that simulates injection molding profiles; integrated Power-Law viscosity models to calculate viscous dissipation, optimizing servo sizing to save \$20 000 across a product line.
- Conducted DC bus electrical load calculations to successfully retrofit a new servo drive to an existing cabinet; collaborated with vendors to validate cabinet thermal clearance and power overhead, preventing costly downtime.
- Authoured and published missing enterprise-level electrical assembly standards, directly resolving 60% non-conformances and clearing critical cabinet release blockers to optimize production DFA.

Tesla | Controls Engineer Intern Jan. 2024 – May 2024
PLC Programming, RFID, DFM, DFA, HIL Testing, Git Markham, Ontario

- Spearheaded an RFID-based tooling monitoring system and custom TwinCAT HMI dashboard, tracking asset lifespan data to reduce maintenance issues by over 90% across Gigafactory lines.
- Commissioned 3 automated production machines by leveraging Beckhoff TwinCAT3 Structured Text to program and synchronize VFDs, sensors, pneumatics, and FANUC robotic arms.
- Debugged and optimized complex field I/O networks across factory automation equipment using multiple industrial protocols including TCP/IP, EtherCAT, and Profinet.
- Led advanced R&D initiatives experimenting with computer vision, time-of-flight lasers, and profilometers, authoring test reports to optimize high-performance device integration.

Moneris | Software Engineer Intern Jun. 2022 – Aug. 2023
React.js, C++, Python, Git, Agile Toronto, Ontario

- Developed a tool boosting productivity for 30+ employees by enabling credit card transaction analytics. Built with JavaScript and React, the tool allows users to customize dashboards and manage widgets seamlessly.
- Improved code efficiency by 50% by implementing Redux and a backend system hosted on Parse, optimizing personal and team tools by implementing reducers, utilizing dispatch, and querying user database properties.
- Collaborated in agile feedback sessions with the Scrum Master, QA team, Architecture team, and other stakeholders to identify and resolve platform issues related to our proprietary payment platform testing tool.

PROJECTS

Autonomous Defense Drone System | *Python, YOLOv8, CAD, RF Modules, Control Algorithms*

- Integrated a custom YOLOv8 drone detection model across 2 cameras, CAD-designed robotic arms, and RF Tx/Rx modules using Python control algorithms to track and intercept unauthorized drone threats to secure airspace.

Autonomous RC Car | *C++, Python, LiDAR, Linux, Microcontrollers*

- Implemented self-driving algorithms in C++ and Python on an NVIDIA Jetson Nano running Linux Ubuntu to process LiDAR and ultrasonic sensor data for obstacle navigation, while coordinating development phases as Scrum Master.

EDUCATION

McMaster University | B.Eng., Electrical Engineering Graduated Apr. 2025